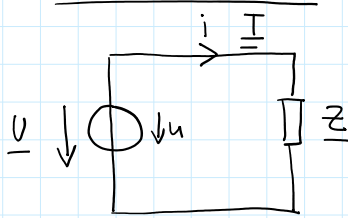


LEISTUNG



$$u = \hat{u} \sin(\omega t + \varphi)$$

$$i = \hat{i} \sin(\omega t)$$

$$P(t) = p = u \cdot i \quad \text{Augenblicksleistung}$$

$$p = u \cdot i = \hat{u} \cdot \hat{i} \cdot \sin(\omega t + \varphi) \cdot \sin(\omega t) = *$$

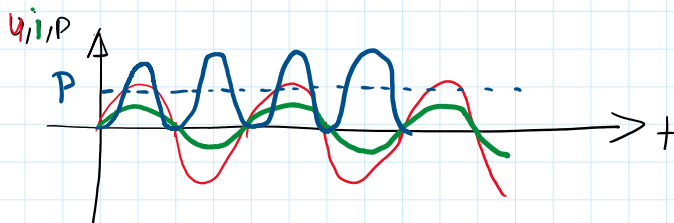
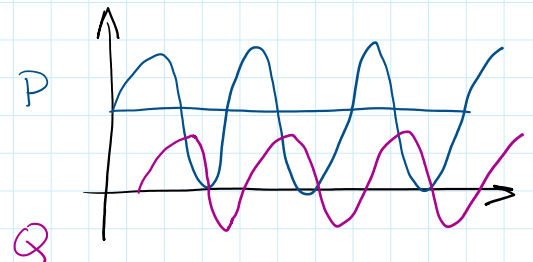
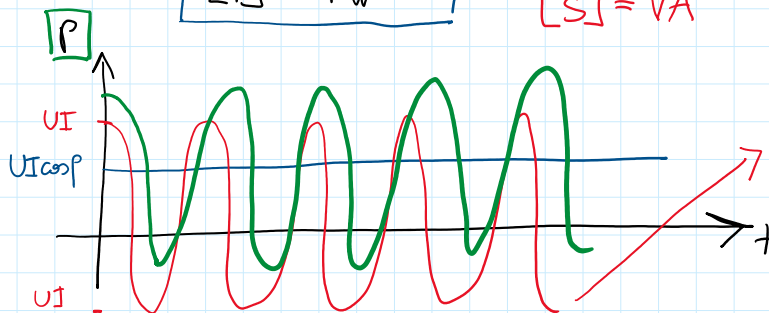
$$\sin \alpha \cdot \sin \beta = \frac{1}{2} [\cos(\alpha - \beta) - \cos(\alpha + \beta)]$$

$$* = \frac{1}{2} \hat{u} \hat{i} [\cos(\varphi) - \cos(2\omega t + \varphi)] =$$

$$= \underbrace{U \cdot I \cdot \cos \varphi}_{(=\text{Gleichwert})} - \underbrace{U \cdot I \cos(2\omega t + \varphi)}_{(=\text{Wechselnde } P(t))}$$

$$\boxed{P \text{ Wirkleistung}} \\ \boxed{[P] = 1 \text{ W}}$$

$$S = U \cdot I \text{ Scheinleistung} \\ [S] = \text{VA}$$



KOMPLEXER LEISTUNG

$$\underline{S} = P + jQ = S \angle \varphi = * (\text{Winkelgeschwindigkeit } 2\omega)$$

$$Q \text{ Blindleistung} \quad [Q] = \text{var}$$

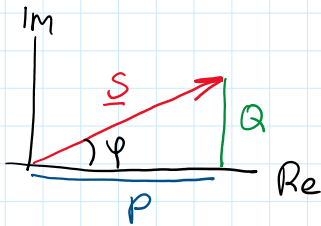
$P > 0$ Verbraucher	$Q > 0$ induktiv
$P < 0$ Erzeuger	$Q < 0$ kapazitiv

$P > 0$ Verbraucher	$Q > 0$ induktiv
$P < 0$ Erzeuger	$Q < 0$ kapazitiv

$$* = S \cdot \cos \varphi + j S \sin \varphi$$

$$\text{Leistungsfaktor } \lambda = \frac{P}{S} = \frac{UI \cos \varphi}{UI} = \cos \varphi$$

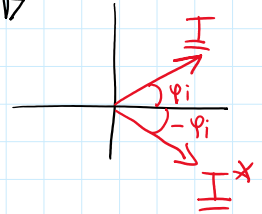
$$-90^\circ < \varphi < 90^\circ \rightarrow \text{Verbraucher}$$



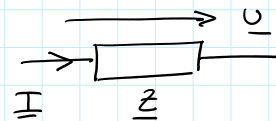
$$\underline{S} = \underline{U} \cdot \underline{I} = U \cdot \underline{I} \angle \varphi_u = U \cdot I \angle \varphi_u + \varphi_i$$

$$\underline{I}^* = I \angle -\varphi_i$$

$$\underline{S} = \underline{U} \cdot \underline{I}^* = U \cdot I \angle \varphi_u - \varphi_i$$



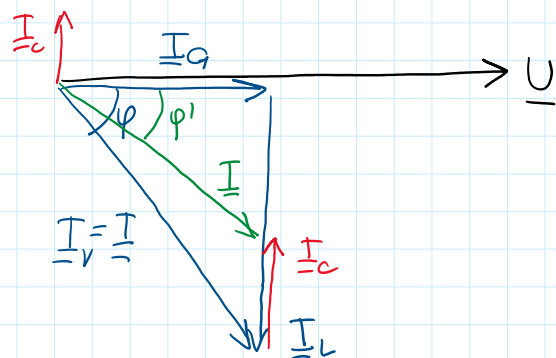
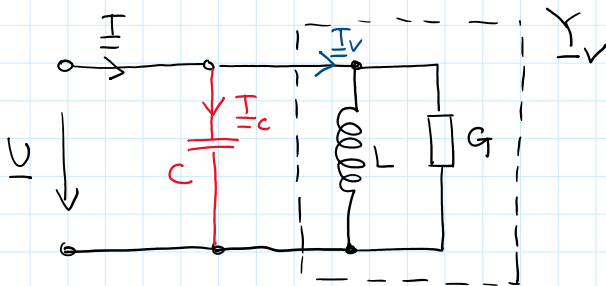
$$\underline{U} = 12,4 \text{ V} \angle 28^\circ \quad \underline{I} = 1,5 \text{ A} \angle -76^\circ$$



$$\begin{aligned} \underline{S} &= \underline{U} \underline{I}^* = 12,4 \text{ V} \angle 28^\circ \cdot 1,5 \text{ A} \angle +76^\circ = 18,6 \text{ VA} \angle 104^\circ = \\ &= \underbrace{-4,5 \text{ W}}_P + j \underbrace{18,05 \text{ var}}_Q \end{aligned}$$

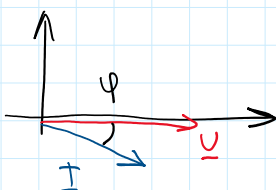
Erzeuger induktiv

BLINDLEISTUNGS KOMPENSATION

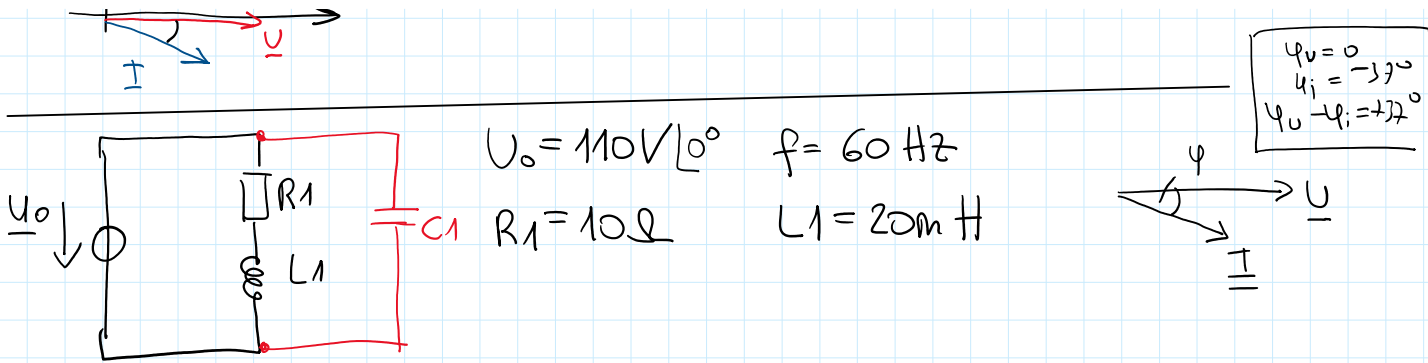


$$\cos \varphi > 0,9 \quad \varphi \approx 25^\circ$$

$$\frac{Q}{P} \sim 48\%$$



$$\begin{aligned} \varphi_u &= 0 \\ \varphi_i &= -25^\circ \end{aligned}$$



a) $\underline{Z} = R_1 + j\omega L_1 = 10 + j7,54 \Omega \Rightarrow \boxed{\varphi = 37,02^\circ}$

$\lambda = \cos \varphi = 0,79$

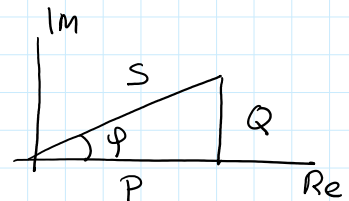
$\underline{I} = \frac{\underline{U}}{\underline{Z}} = \frac{110V \angle 0^\circ}{10 + j7,54 \Omega} = 7,03 - 5,29j \text{ A} = \underline{8,8A \angle -37^\circ}$

$P = U \cdot I \cdot \cos \varphi = 771,6 \text{ W}$ $Q = U \cdot I \cdot \sin \varphi = 581,2 \text{ var}$

b) Kompensation $C_1 \rightarrow \lambda = 0,98$

$\cos \varphi = 0,98 \rightarrow \varphi = 11,48^\circ$

$\tan \varphi = \frac{Q}{P}$ $Q = \tan \varphi \cdot P = 156,7 \text{ var}$



$\Delta Q = -(581,2 - 156,7) \text{ var} = -424 \text{ var}$

$Q_C = -U^2 \omega C_1 \rightarrow C_1 = 9,317 \cdot 10^{-5} \text{ F} = \underline{93,17 \mu\text{F}}$

c) Absinken von $\underline{I}$ $\underline{Z}_{C_1} = \frac{1}{j\omega C_1} = -j \frac{1}{\omega C_1}$

$\underline{Z}_n = \frac{\underline{Z} \cdot \underline{Z}_c}{\underline{Z} + \underline{Z}_c} = 15,06 + 3,06j \Omega = \underline{15,37 \Omega \angle 11,48^\circ}$

$\underline{I}_n = \frac{\underline{U}}{\underline{Z}_n} = \frac{110V}{15,37 \angle 11,48^\circ \Omega} = 7,16A \angle -11,48^\circ \Rightarrow -18,5\% \text{ Reduktion}$

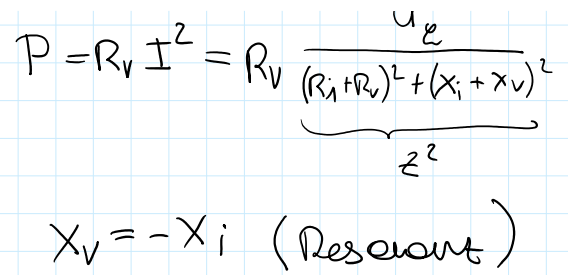
1 € = 100 ct = 10ct × 10ct =

= 0,1€ × 0,1€ = 0,01€ = 1ct

LEISTUNGS ANPASSUNG

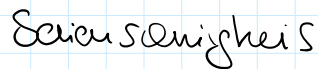


$P = R_L I^2 = R_L \frac{U_g^2}{(R_i + R_L)^2 + (X_i + X_v)^2}$



$$R_v = R_i \quad X_v = -X_i \quad \underline{Z}_v = R_i - jX_i = \underline{Z}_i^*$$

RESONANZ

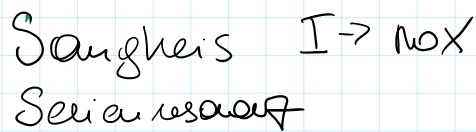


$$\omega L = \frac{1}{\omega C}$$

$$\omega^2 = \frac{1}{LC}$$

$$\omega = \frac{1}{\sqrt{L C}}$$

$$f = \frac{1}{2\pi\sqrt{LC}}$$

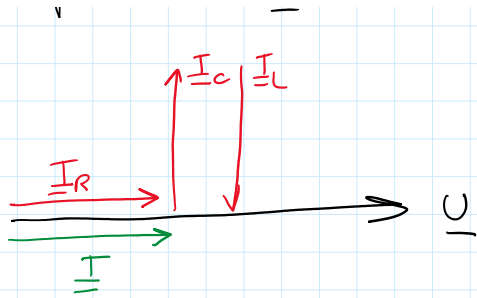


$$\underline{Y} = \frac{1}{R} - j\frac{1}{\omega L} + j\omega C = \frac{1}{R} + j(\omega C - \frac{1}{\omega L})$$

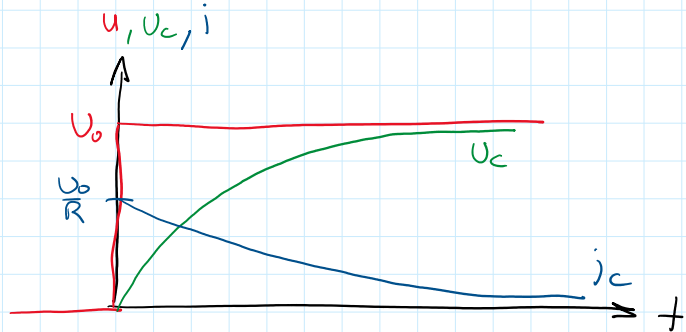
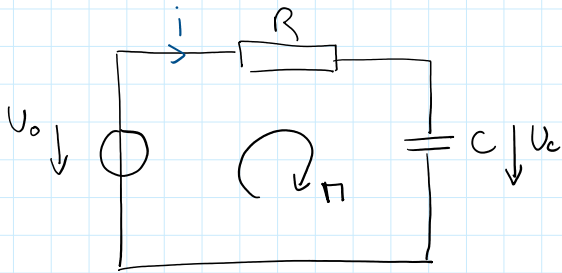
$$f = \frac{1}{2\pi \sqrt{LC}}$$

Spankeis I $\rightarrow$ Min

$$\uparrow \underline{I_C} \mid \underline{I_L}$$



TRANSIENTENVERHALTEN



$$-U_0 + R \cdot i + U_C = 0$$

$$i_C = C \cdot \frac{dU_C}{dt}$$

$$-U_0 + R \cdot C \cdot \frac{dU_C}{dt} + U_C = 0$$

$$\boxed{\frac{dU_C}{dt} + \frac{1}{RC} U_C = \frac{1}{RC} U_0}$$

$$U_C = U_C^h + U_C^p$$

h... homogen
p... partikulär

1) homogene Lösung

$$\frac{dU_C}{dt} + \frac{1}{RC} U_C = 0$$

$$\frac{dU_C}{dt} = -\frac{1}{RC} U_C$$

$$\frac{1}{U_C} dU_C = -\frac{1}{RC} dt \xrightarrow{\int} \ln(U_C) = -\frac{1}{RC} t + \ln(k)$$

$$U_C^h = k \cdot e^{-\frac{t}{RC}} = k e^{-\frac{t}{\tau}}$$

$$\tau = R \cdot C \quad \text{Zeitkonstante}$$

2) partikuläre Lösung

$$U_C^p = U_0$$

$$U_C = k \cdot e^{-\frac{t}{\tau}} + U_0$$

3) Anfangsbedingung

$$t=0 \rightarrow U_C=0$$

$$U_c(t=0) = k + U_0 = 0 \Rightarrow k = -U_0$$

4) Lösung

$$U_c = -U_0 e^{-\frac{t}{\tau}} + U_0 = U_0(1 - e^{-\frac{t}{\tau}})$$

$$i = C \cdot \frac{dU_c}{dt} = C \cdot \frac{d}{dt} (U_0 - U_0 e^{-\frac{t}{\tau}}) =$$

$$= C(-U_0 \cdot (-\frac{1}{\tau}) \cdot e^{-\frac{t}{\tau}}) = \frac{U_0}{R} e^{-\frac{t}{\tau}} = \frac{U_0}{R} e^{-\frac{t}{\tau}}$$

$$P_c = i \cdot U_c = \frac{U_0}{R} e^{-\frac{t}{\tau}} \cdot U_0(1 - e^{-\frac{t}{\tau}}) = \frac{U_0^2}{R} (e^{-\frac{t}{\tau}} - e^{-\frac{2t}{\tau}})$$

$$W_c = \int_0^{\infty} P_c dt = \int_0^{\infty} \frac{U_0^2}{R} (e^{-\frac{t}{\tau}} - e^{-\frac{2t}{\tau}}) dt =$$

$$= \frac{U_0^2}{R} \left[\frac{e^{-\frac{t}{\tau}}}{-\frac{1}{\tau}} - \frac{e^{-\frac{2t}{\tau}}}{-\frac{2}{\tau}} \right]_0^{\infty} = \frac{U_0^2}{R} \left[-\tau e^{-\frac{t}{\tau}} + \frac{\tau}{2} e^{-\frac{2t}{\tau}} \right]_0^{\infty} =$$

$$= \frac{U_0^2}{R} \left[0 + 0 + \tau - \frac{\tau}{2} \right] = \frac{U_0^2 \tau}{2R} = \frac{U_0^2 C}{2}$$

